SV16 BOM - rebuild step 1 (value-level changes applied)

Source: SBSV-16.kicad_pcb (uploaded), PCB_COMPONENTS.pdf (spec), cart screenshots. Status column: BUY = not in cart, OK = covered by cart, UNDECIDED = needs a decision, PENDING = PCB footprint/value change still to do. Shared parts are listed once in the cart column.

Spec part	Board value (step1)	Footprint (board)	Bought (cart)	Status	
LFE5U-12F-6TG144C	LFE5U-12F-6TG144C	LQFP-144	0	BUY (FPGA)	
W25Q64JVSSIQ	W25Q64JVSSIQ	SOIC-8 5.3x6.2	1	OK (have 1 of 1)	
W25Q64JVSSIQ	W25Q64JVSSIQ	SOIC-8 5.3x6.2	1	IN CART (qty 1); with 1 owned = 2 total. Covered.	
Active 25 MHz XO 3.2x2.5 4-pad (ECS-3225MV-250)	25 MHz ACTIVE XO	ASE-4Pin 3.2x2.5 (verify pads vs ECS land pattern)	0	BUY ECS-3225MV-250 (3 passive crystals bought - NOT usable here)	
12 MHz crystal HC49/U surface mount (passive 2-pin, OK for CH340G)	12 MHz	HC49-4H (THT) - change to SMD HC49/US	0	BUY 12 MHz SMD crystal; PENDING footprint change to SMD HC49	
CH340G	CH340G	SOIC-16	1	OK	
AP62300TWU-7	AP62300TWU-7	SOT-23-6	1	OK	
MP1584EN	MP1584EN	SOIC-8-1EP	1	OK	
LP5907MFX-2.5	LP5907MFX-2.5	SOT-23-5	1	OK	
USBLC6-2SC6	USBLC6-2SC6	SOT-23-6	4	OK	
2N7002	2N7002	SOT-23	7	OK	
MMBT3904	MMBT3904 (changed)	SOT-23	9	OK	
MMBT3906	MMBT3906 (changed)	SOT-23	13	OK	
LED green 0603	LED green	0603	7	OK	
LED red 0603 (DNP by default per spec)	LED red	0603	6	OK - mark DNP on board (not done yet)	
LED red 0603	LED red	0603	6	OK	
SS34	SS34	SMA	2	BUY 1 more (need 3 total with D9/D11)	
SS34	SS34	SMA	2	(see D8)	
SS34 (1V1 catch diode)	SS34	SMA	2	(see D8)	
10 uH CD54 >=1.5A	10 uH	SWPA6045S footprint (spec: MWSA0518) - footprint NOT changed	2 (CD54 10uH)	OK value; footprint change pending	
10 uH CD54 >=1.5A	10 uH	SWPA6045S footprint (spec: MWSA0518) - footprint NOT changed	2 (CD54 10uH)	OK value; footprint change pending	
600R @100MHz 0805 (MPZ2012S601ATD25)	600R @100MHz	0805	2	IN CART (TDK MPZ2012S601ATD25	qty 2); footprint change 0603
100uF 25V D-case tant	100uF 25V tant	CP_EIA-7343-31	1	OK	

Spec part	Board value (step1)	Footprint (board)	Bought (cart)	Status	
22uF 63V electrolytic	22uF 63V electrolytic	CP_Elec_6.3x5.4	22	OK	
470uF 25V electrolytic (CHANGED from 16V)	470uF 25V	CP_Elec_8x10.5	0	BUY 470uF 25V (16V stock not used for C23; derating 75% too high)	
220uF 25V hybrid	220uF 25V hybrid	CP_Elec_8x10.5	11	OK	
10uF 0805 ceramic (spec) -> 10uF 35V electrolytic (decision)	10uF	C_0805 - footprint NOT changed	11 (10uF 35V electrolytic)	footprint change pending (CP_Elec_4x5.4)	
10uF 0805 ceramic (spec) -> 10uF 35V electrolytic (decision)	10uF	C_0805 - footprint NOT changed	11	footprint change pending	
10uF electrolytic (decision)	10uF 35V electrolytic	CP_Elec_4x5.4	11	OK	
22uF electrolytic (decision)	22uF 63V electrolytic	CP_Elec_6.3x5.4	22	OK	
22uF electrolytic (decision)	22uF 63V electrolytic	CP_Elec_6.3x5.4	22	OK	
10uF electrolytic (decision)	10uF 35V electrolytic	CP_Elec_4x5.4	11	OK	
470nF 0603 (spec)	470nF	C_0805 - footprint NOT changed	20 (470nF 0603)	footprint change pending (0603)	
CH340G V3 decoupling (U4 pin 4 to GND), in schematic	100nF	0603	40	KEEP (in schematic, required for CH340G)	
COMP capacitor (U6_COMP_RC to GND). Schematic says 100n, board 100pF	100pF (recommended)	0603	0	BUY/CHECK 100pF 0603 C0G. Change schematic C35 from 100n to 100pF.	
22pF C0G 0805 50V 5% (YAGEO CC0805JRNPO9BN220)	22pF	0603 - change to 0805	0	BUY 22 pF 0805 C0G; PENDING footprint change 0603->0805	
22pF C0G 0805 50V 5% (YAGEO CC0805JRNPO9BN220)	22pF	0603 - change to 0805	0	BUY (see C36); PENDING footprint change 0603->0805	
100nF 0603	100nF	0603	40 (shared)	OK	
100nF 0603	100nF	0603	(shared)	OK	
470nF 0603	470nF	0603	(shared)	OK	
100nF 0603	100nF	0603	(shared)	OK	
100nF 0603	100nF	0603	(shared)	OK	
10k 0603 5% (YAGEO RC0603JR-0710KL)	10k	0603	0	BUY 10k 0603 5% (Yageo RC0603JR-0710KL); in cart (qty 110)	
4.7k 0805	4.7k (changed)	0805	(4.7k 0805 pack)	OK	
4.7k 0805 (from bought pack)	4.7k	0603 - change to 0805	(4.7k 0805 pack)	PENDING footprint change 0603->0805; use bought 4.7k 0805 pack	
1k 1206 (from bought pack)	1k	0603 - change to 1206	(1k 1206 pack)	PENDING footprint change 0603->1206; use bought 1k 1206 pack	
100R 1206 (spec) - board 1206	100R (changed)	1206	(100R 0805 pack bought)	footprint change to 0805 pending (bought 0805)	

Spec part	Board value (step1)	Footprint (board)	Bought (cart)	Status	
10k 0603 5% (YAGEO RC0603JR-0710KL)	10k	0603	0	BUY 10k 0603 5% (Yageo RC0603JR-0710KL), 7 pieces	
0R	0R (changed)	0603	0	solder-bridge option (no purchase needed)	
4.7k 0805 (from bought pack)	4.7k	0603 - change to 0805	(4.7k 0805 pack)	PENDING footprint change 0603->0805; use bought 4.7k 0805 pack	
10k 0603 5% (YAGEO RC0603JR-0710KL)	10k	0603	0	BUY 10k 0603 5% (Yageo RC0603JR-0710KL)	
470R 0805	470R (changed)	0805	(470R 0805 pack)	OK	
1k 1206	1k (changed)	1206	1 (1k 1206 pack)	OK	
10k 0603 5% (YAGEO RC0603JR-0710KL)	10k	0603	0	BUY 10k 0603 5% (Yageo RC0603JR-0710KL)	
1k 1206	1k (changed)	1206	(1k 1206 pack)	OK	
1k 1206	1k (changed)	1206	(1k 1206 pack)	OK	
5.1k 0805	5.1k (changed)	0805	(5.1k 0805 pack)	OK	
5.1k 0805	5.1k (changed)	0805	(5.1k 0805 pack)	OK	
12.4k 0402	12.4k (changed)	0402	30	OK	
33k 1206	33k (changed)	1206	(33k 1206 pack)	OK	
100k 0603	100k (changed)	0603	11 (thin film)	OK	
100k 0603	100k (changed)	0603	(shared)	OK	
100k 0603	100k (changed)	0603	(shared)	OK	
LED D7 current limit (3V3 to D7 red LED), 1k in schematic	1k	0603	0	CHECK stock for 1k 0603, buy if none. Board had no value; schematic says 1k.	
COMP resistor (U6 COMP to U6_COMP_RC). Schematic says 10k, board 100k, calculation 150k	150k (recommended)	0603	0	BUY 150k 0603. Change schematic R45 from 10k to 150k.	
33k 1206	33k (changed)	1206	(shared)	OK	
10k 0603 1% (YAGEO RC0603JR-0710KL, 5%)	10k	1206 - change to 0603	0	BUY 10k 0603 5% (Yageo RC0603JR-0710KL); footprint change 1206->0603	
AP62300 EN divider (value not given in spec)	100k	0603	(shared)	KEEP board value (not in spec, purpose unverified) - report	
AP62300 EN divider (value not given in spec)	33k	1206	(shared)	KEEP board value (not in spec, purpose unverified) - report	
Standard 2.54 mm straight male pin header, 1x10 (most available). Footprint unchanged.	JTAG	2.54mm 1x10	0	USER HAS (not in screenshots) - confirm	

Spec part	Board value (step1)	Footprint (board)	Bought (cart)	Status	
Spec: 1.27 mm 2x5 box header. Less common; leave unpopulated and use J1 (spec option).	JTAG-1.27	1.27mm 2x5	0	OPTIONAL	
Standard 2.54 mm straight male pin header, 2x10 (most available). Footprint unchanged.	EXP-A	2.54mm 2x10	0	USER HAS (not in screenshots) - confirm	
Standard 2.54 mm straight male pin header, 2x10 (most available). Footprint unchanged.	EXP-B	2.54mm 2x10	0	USER HAS (not in screenshots) - confirm	
Standard 2.54 mm straight male pin header, 1x10 (most available). Footprint unchanged.	EXP-C	2.54mm 1x10	0	USER HAS (not in screenshots) - confirm	
Shrouded keyed 1x6 2.54 mm header (motor connector)	MOTOR	2.54mm 1x6	0	BUY shrouded keyed 1x6 2.54 mm (or confirm you have one)	
Standard 2.54 mm straight male pin header, 2x25 (footprint only, not fitted).	absent	-	0	ADD footprint (not fitted)	
USB-C 16-pin HRO	USB-C	HRO TYPE-C-31-M-12	4	OK	
Barrel jack 5.5/2.1	7-12 V	CUI PJ-102AH	0	USER HAS (not in screenshots) - confirm	
Standard 2.54 mm straight male pin header, 1x4 (most available). Footprint unchanged.	CONSOLE	2.54mm 1x4	0	USER HAS (not in screenshots) - confirm	
Standard 2.54 mm straight male pin header, 1x2, with shunt (most available). Footprint unchanged.	DTR ENABLE	1x02	0	USER HAS (not in screenshots) - confirm	
6x6 tact	RESET	SW_PUSH_6mm	0	USER HAS (not in screenshots) - confirm	
6x6 tact	RECONFIG	SW_PUSH_6mm	0	USER HAS (not in screenshots) - confirm	
1 mm pads	OK	TestPoint	-	OK	
M3 mount holes	OK	-	-	OK	